



Department of Telecommunications
Telecommunication Engineering Centre
K. L. Bhawan, Janpath, New Delhi – 110001
(Convergence & Broadcasting Division)



No: 4-16/2018-C&B/TEC/NWG-9

Dated: 16 Oct 2023

Sub: Minutes of the 11th Meeting of National Working Group-9 (NWG-9) regarding "Broadband Cable and TV", held on 12.10.2023.

The 11th meeting of the National Working Group-9 (NWG-9) corresponding to the ITU-T Study Group 9 (SG9), regarding "Audiovisual content transmission and integrated broadband cable networks/ Broadband Cable and TV", was held on 12.10.2023 under the chairmanship of Sh. Avinash Agarwal, DDG (Convergence & Broadcasting), TEC through video-conferencing. List of participants is placed at Annexure-I.

2. Sh. Harsh Sharma, Convener NWG-9 & Director (C&B) TEC, welcomed all the participants and provided a summary of the previous meeting held on 21.09.2023.
3. Sh. Avinash Agarwal, Chairman NWG-9 & DDG (C&B) TEC, in his opening remarks, briefed about the forthcoming meeting of ITU-T Study Group 9 (SG9) scheduled in Bogota (Columbia) from 14-23 Nov 2023. He conveyed that ITU-T has issued the detailed meeting information vide TSB Collective Letter dated 13.09.2023. The last date for uploading contributions on the ITU-T portal is 01.11.2023. Details may be referred at:

<https://www.itu.int/en/ITU-T/studygroups/2022-2024/09/Pages/default.aspx>

He briefed about all the Questions and work items currently under study in SG9. Presently, SG9 is working on total 11 questions in the current truncated study period of 2022-2024. He also welcomed the new members who joined the NWG9 meeting for the first time.

4. Sh. Avinash Agarwal, DDG(C&B) TEC informed that India, through previous contributions, was successful in release of one ITU Recommendation and one ITU Technical Report, both related to the work on Conditional Access Systems (CAS) done by TRAI and C&B Division of TEC. Details are as follows:
 - a. **Recommendation ITU-T J.1036** 'Factual subscriber-base reporting and protected content delivery in conditional access system – Requirements';
 - b. **ITU-T Technical report TR.FSR** 'Factual subscriber-base reporting and protected content delivery in a conditional access system (CAS).'
5. Following contributions received from members were discussed:
 - a. Q6/9: New work item on "Functional requirements for 8k cable set-top box (STB) with Virtual reality (VR) services": Dr. P. C. Jain, IAFI and Ms. Meenakshi Singhvi, Prasar Bharati presented this contribution related to initiating a new work item in Q6/9 for a new recommendation on an 8k cable set-top box integrated with virtual reality service using WiFi-7 technology. It was informed that the rapid development of UHD video business and virtual reality industry has put forward new requirement for the integration of set-top box and virtual reality equipment. Sh. Paritosh Saha and Sh. Rajeev Khattar provided some useful inputs. Sh. Avinash Agarwal advised to prepare the contribution in proper format including the justification form and the baseline text.
 - b. Q6/9: New work item on "Set Top Box as Home Gateway or Smart Set Top Box": Ms. Meenakshi Singhvi presented this contribution informing that it proposes to formulate the

specification of a smart set-top box that will also serve as an aggregator for IoT based smart home devices. This is proposed keeping in view the recent technological developments. Sh. Paritosh Saha, Sh. Rajeev Khattar, Sh. Ramesh DS and other members shared their views regarding this proposal. Sh. Paritosh Saha informed that details about a similar product could be accessed at http://www.kaonmedia.com/eng/product/AI_STB.asp. Sh. Avinash Agarwal advised to check the availability of sufficient technical content before taking it up as a new work item. He also observed that presently the content looks appropriate for a Technical Report instead of a Recommendation. It was felt that the contribution may be updated and may be considered during the next NWG9 meeting.

- c. Q2/9: J.FSR-TEST 'Factual subscriber-base reporting and protected content delivery in conditional access system – Test Methods': Sh. Avinash Agarwal briefed that J.FSR-TEST is the 3rd work item created by India in this series and was created during the previous SG9 meeting. It provides clause-by-clause test procedures and expected results for each clause of ITU-T recommendation J.1036 and is fully based on the Test guide prepared by the C&B Division of TEC. This contribution provides the details of the remaining clauses and could be submitted for consent in the forthcoming SG9 meeting.
 - d. Q11/9: JSTR.LCAP (Technical Report) 'Technical advances, challenges, and best practices in live captioning': Sh. Avinash Agarwal presented the contribution for updating the baseline text of this work item w.r.t. the last output document (i.e. TD342). Sh. Shubhanshu Gupta, CDAC also briefed about the additions done. Sh. Avinash Agarwal requested to Sh. Rajeev Khattar and BIF to examine the six documents referred by ITU-R SG6 and provide inputs for this work item.
6. Sh. Avinash Agarwal suggested that we should also submit a contribution on our final version of the updated recommendation J.298 regarding hybrid set-top box.
 7. After detailed deliberations, the NWG9 recommended the following contributions for submission to ITU-T SG9 and decided that their final versions would be reviewed during the next NWG9 meeting:

S.N.	Question	Topic	Contributors
1	Q2/9	Updated baseline text of J.FSR-TEST 'Factual subscriber-base reporting and protected content delivery in conditional access system' for consent	Avinash Agarwal, TEC Harsh Sharma, TEC
2	Q6/9	Proposed A.1 justification for a new ITU-T recommendation on "Functional requirements for 8k cable set-top box (STB) with Virtual reality (VR) services"	Avinash Agarwal, TEC Meenakshi Singhvi, Prasar Bharati Prem Chandra Jain, IAFI
3	Q6/9	Proposed baseline text for the new ITU-T recommendation on "Functional requirements for 8k cable set-top box (STB) with Virtual reality (VR) services"	Avinash Agarwal, TEC Meenakshi Singhvi, Prasar Bharati Prem Chandra Jain, IAFI
4	Q6/9	Updated baseline text of recommendation J.298 regarding hybrid set-top box for consent	Meenakshi Singhvi, Prasar Bharati Rajiv Khattar Paritosh Shah Ramesh DS
5	Q11/9	JSTR.LCAP (Technical Report) 'Technical advances, challenges, and best practices in live captioning	Avinash Agarwal, TEC Shubhanshu Gupta, CDAC Rajiv Khattar

8. Chairman NWG9 conveyed that in the upcoming meeting of SG9, there will be special sessions regarding preparation of WTSA-2024. All NWG members were requested to examine the current

profile/responsibilities of SG9 and terms of references of the various Questions and identify areas/topics that required to be updated.

9. Chairman NWG9 conveyed that NWG9 members who plan to attend the SG9 meeting, either physically or virtually, should have the TIES account. Members may send a formal request to TEC alongwith the requisite form and undertaking for creation of their TIES account.
10. NWG9 decided to holding the next meeting in next week on 20th OCT 2023 (forenoon session) to review and finalise the contributions for the forthcoming SG9 meeting.



(Harsh Sharma)

Convener, NWG-9 & Director, (C&B), TEC

Encl.: Annexure-I

To,

All members of the NWG-9 and all the Participants.

Copy for kind information to:

1. Advisor/Sr. DDG TEC, New Delhi
2. Chairman NWG-9 & DDG (C&B), TEC, New Delhi.

List of Participants:

1. Sh. Avinash Agarwal, Chairman NWG-9 & DDG (C&B), TEC.
2. Ms. Meenakshi Singhvi, Director, AIR & Doordarshan, Prasar Bharati Mumbai
3. Sh. Satbir Singh, M/o I&B
4. Other representative from M/o I&B
5. Sh. Shubhanshu Gupta, C-DAC
6. Ms. Seema Bhagwan Kawale, SAKEC, Mumbai
7. Ms. Uma Raj, SAKEC, Mumbai
8. Sh. Rajiv Khattar, Researcher & Industry specialist
9. Sh. Paritosh Saha, Disney-Hotstar
10. Sh. Ramesh DS, M/s Exalto Communication Ltd.
11. Dr. Prem Chand Jain, IAFI
12. Sh. Gaurav Aggarwal, Vantiva
13. Sh. Manav
14. Sh. Harsh Sharma, Convener, NWG-9 & Director (C&B), TEC.
